
Algorithm 1: L-BFGS Two-Loop Recursion for $r = H_m q$

Input: q , stored pairs $\{(s_i, y_i)\}_{i=0}^{m-1}$, initial matrix H_0

Output: $r = H_m q$

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1 for  $i = m - 1, m - 2, \dots, 0$  do
2   |    $\rho_i \leftarrow 1/(y_i^\top s_i)$ 
3   |    $\alpha_i \leftarrow \rho_i s_i^\top q$ 
4   |    $q \leftarrow q - \alpha_i y_i$ 
5 end
6  $r \leftarrow H_0 q$ 
7 for  $i = 0, \dots, m - 1$  do
8   |    $\beta_i \leftarrow \rho_i y_i^\top r$ 
9   |    $r \leftarrow r + s_i(\alpha_i - \beta_i)$ 
10 end
11 return  $r$ 
```
